

Research Statement

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Environmental quality depends on incentives. My research studies how changes in incentives reshape resource use, environmental quality, and social welfare. I focus on settings where individual decisions generate environmental externalities that markets fail to internalize, including agriculture, forestry, transportation, and energy.

My research has two complementary themes. The first examines how resource-dependent households and firms manage environmental resources under changing incentives. The second studies how transportation systems generate externalities and how policies and shocks reshape travel behavior. Together, these projects investigate how economic decisions interact with environmental constraints and public policy.

1. Environmental and Natural Resource Management

Agriculture and forestry depend on natural resources whose environmental values are often not reflected in market incentives. This research examines how resource-dependent households respond to changing incentives and how these responses affect production, environmental outcomes, and risk.

- **Farm Household Economics** Liou et al. ([2025a](#)), Lee et al. ([2025a](#)), Chang and Liou ([2026](#)), and Lee et al. ([2025b](#)) study how agricultural and forest households manage production decisions, environmental resources, and economic risk. The studies show that renewable-energy adoption can complement agricultural production by increasing farm sales and labor demand. They also show that food traceability adoption reduces chemical input use and groundwater extraction while improving farm revenue and market access. In forest management, greater tree diversity involves an economic trade-off: it reduces expected income but lowers income volatility, highlighting biodiversity as natural capital that provides risk-reduction benefits.
- **Climate Policy** Liou et al. ([2025b](#)) extends this research agenda to international climate policy by examining how carbon border adjustment policies affect incentives for cleaner pro-

duction and reshape global production decisions.

2. Transportation and Environmental Externalities

Transportation generates substantial externalities through accidents, congestion, pollution, and carbon emissions. This research examines how transportation policies and large-scale shocks affect travel behavior, environmental outcomes, and social welfare.

- **Large-scale Event and Road Safety** Liou and Chang (2026a), Liou and Chang (2026b), and Chang et al. (2024) study how institutional changes and large-scale events affect traffic risks and environmental outcomes. The studies show that traffic regulation reform can reduce accidents, injuries, and medical costs, while large-scale events increase traffic risks and pollution primarily through changes in travel demand.
- **Travel Demand and COVID-19** Liou et al. (2025c), Chang et al. (2021), Liou et al. (2024), and Liou et al. (2026) examine how transportation systems and local economies respond to policy changes and large-scale shocks. The studies show that transit pricing can increase mobility while reducing carbon emissions, while COVID-19 altered travel behavior and slowed economic recovery through changes in perceived risk and consumer decisions.

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